Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

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1 Introduction

Under the European Union's standard value-added tax (VAT), the seller charges tax on the invoice and remits it to the government, known as seller remittance. For many cross-border business-to-business supplies, the reverse charge mechanism instead shifts that obligation to the buyer, known as buyer remittance. Under reverse charge, the seller invoices without VAT, and the buyer accounts for the tax. A similar distinction appears in the United States between sales tax, collected and remitted by the seller, and use tax, which the buyer self-assesses on out-of-state purchases when the seller has no nexus.[1]

When the transaction is final and the good is kept, these remittance rules are economically equivalent. The treasury receives the same revenue, and which party remits is only an administrative detail. A return changes that picture, because the tax paid to the government must itself be unwound. Reclaiming that tax typically requires filing and can involve delay or incomplete recovery, so the remitter's effective tax refund is smaller than the statutory amount. We take the extreme case in which the remitter recovers none of the tax from the government on a returned unit—whether the seller or the buyer remitted.[2] Even so, the private refund between buyer and seller still differs by regime. Under seller remittance the seller refunds the full tax-inclusive checkout price, while under buyer remittance the seller refunds only the pre-tax price and the buyer does not recover the tax. The keep-or-return decision and the seller's effective margin therefore depend on which party remits.

Most incidence and pass-through analysis abstracts from this difference. It assumes remittance neutrality and treats a return as undoing the purchase, so pass-through, incidence, and deadweight loss are read off demand and supply alone. That abstraction is hard to justify in practice. About one in five online purchases is returned,[3] and a return is not a free reversal of the transaction. Never buying and buying then returning leave the same final allocation—the consumer does not keep the good—yet a returned purchase incurs shipping, hassle, and handling that a non-purchase

[1] Since the 2018 *Wayfair* decision, nearly every state with a sales tax has adopted "marketplace facilitator" laws that require platforms such as Amazon, eBay, and Etsy to collect and remit tax on behalf of their third-party sellers. As a result, the platform, not the independent seller, has become the statutory remittance agent for the vast majority of U.S. e-commerce transactions. This shift has reduced the frequency with which consumers must self-assess use tax, while concentrating remittance responsibility in a small number of large platforms.

[2] Partial reclaim from the government would lie between full recovery of τ and this zero-reclaim benchmark. We use the extreme to put both remittance regimes on the same footing with respect to treasury friction.

[3] National Retail Federation and Happy Returns, *2025 Retail Returns Landscape*, October 15, 2025, `https://nrf.com/research/2025-retail-returns-landscape`. The report estimates that 19.3 % of online sales will be returned in 2025.

avoids. Demand and supply alone therefore omit a margin between purchase and final ownership that matters for welfare.

What objects then replace the classical demand and supply curves, and how misleading is tax analysis that omits returns? We study these questions under perfect competition with product returns and seller handling costs. We characterize pass-through, incidence, and welfare losses with returns, compare them to the no-return benchmark, and identify where omitting the return margin changes conclusions.

Even with a return margin, surplus continues to rest on a small set of market aggregates. What changes is which aggregates matter and at which prices they are evaluated. Consumer surplus depends on only two aggregates: gross demand and the return rate. Heterogeneity in tastes, match values, and return costs enters welfare solely through these aggregates. Producer surplus retains the usual geometry—area to the left of the supply curve—but is evaluated at the effective margin, list price minus expected refunds and handling, rather than at list price.

Because consumers pay the list price while firms respond to the effective margin, how that margin moves with the list price governs pass-through from cost shocks into list prices. The response of the effective margin to the list price reflects two forces. First, a higher price directly raises revenue on kept units. Second, the return rate itself may change with price. When the behavioral response dominates—consumers become more likely to return as price rises—the effective margin can actually fall with the list price. This creates a downward-sloping supply curve in list-price space. Because the refund after a return differs by remittance regime, seller and buyer remittance generate different effective margins and equilibria, and therefore different pass-through and incidence.

We show that pass-through of a one-dollar production-cost shock equals that of a seller-remitted unit tax, but differs under buyer remittance. The ratio of handling to production-cost pass-through equals the aggregate return rate. When the behavioral response of returns to price is strong enough that the effective margin falls in the list price, buyer- and seller-remitted pass-through carry opposite signs—a reversal of the classical tax-neutrality intuition. Which regime yields a negative pass-through depends on the relative magnitudes of the supply and demand elasticities. Intuitively, negative pass-through arises when the list-price supply curve slopes downward—a higher price lowers the effective margin and hence quantity supplied—and more steeply than demand. In this region, a cost increase creates excess demand, and the equilibrium price must fall so that supply expands by more than demand.

Despite pass-through of a cost shock potentially being negative, the incidence ratio—the burden on consumers relative to producers—is always positive and identical across all four shocks: production cost, handling cost, and both seller- and buyer-remitted taxes. Consumer and producer surplus therefore always move together, as in the classical model without returns.

The common direction of those surpluses nevertheless hinges on the sign of pass-through and on remittance. When cost pass-through is negative, production cost, handling cost, and seller tax shocks raise both surpluses, yielding a Pareto improvement, while buyer-remitted taxes lower both. In contrast, when pass-through is positive, all four shocks reduce both surpluses. Remittance therefore matters for the welfare direction of a tax when pass-through is negative.

These welfare effects also reshape first-order changes in total private surplus. We decompose the change from a cost or tax increase into an ex-ante piece on the purchase margin and an ex-post piece on the keep-or-return margin. For a production-cost or seller-tax shock, the ex-ante piece is the classical rectangle from fewer units sold. The ex-post piece captures how the price change reallocates units between kept and returned states, with each additional return wasting the list price plus handling cost. Without returns, private surplus falls by exactly the market quantity. With returns, that fall is joined by an ex-post adjustment that the classical no-return formula omits entirely. The same two pieces appear for the other shocks after a simple rescaling. A handling-cost shock scales both by the return rate, and under buyer remittance both are scaled by the response of the effective margin to list price. For taxes we then add treasury revenue $G = \tau Q(\tau)$ back to obtain social surplus, so the first-order deadweight loss of a seller-remitted tax is only the ex-post return-margin term.

Whether the omitted ex-post term adds to or offsets that market-quantity fall hinges on how the return rate responds to the price change. When pass-through is positive and the return rate rises with price, more units are returned and handling costs rise, so the classical formula underestimates the welfare loss. When the return rate instead falls with price, the ex-post term is a gain that partially offsets the ex-ante loss, so the classical formula overestimates it. When pass-through is negative, that same omitted ex-post gain is what delivers the Pareto improvement above. An analyst who ignores returns then not only mismeasures the magnitude but gets the wrong sign.

To sign that bias, we provide a simple diagnostic, from the return-rate elasticity and the ratio of handling to list price, for when the classical no-return pass-through formula over- or underestimates the correct returns-adjusted pass-through. Parallel conditions in the return-rate, supply, and demand elasticities sign when the classical welfare formula over- or understates the true first-order welfare change, or reverses its sign. Those welfare-improving outcomes stand in sharp contrast to the classical no-return benchmark, where pass-through is always positive and such gains are impossible.

2 Setting

A homogeneous product sells at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility of sending the good back, and in the post-purchase match distribution G_i over match quality u. After buying, type i draws $u \sim G_i$ and keeps the item when $u - p \geq -\kappa_i$, earning $u - p$, and returns it otherwise, earning $-\kappa_i$.

Expected surplus from buying at price p is therefore

$$cs_i(p) = \mathbb{E}_{u \sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

On the purchase margin, all of this consumer-side heterogeneity collapses into willingness to pay v_i, defined by $cs_i(v_i) = 0$: type i buys if and only if $p \leq v_i$. Across types, v_i has distribution F with density f, so gross demand is $D(p) = 1 - F(p)$. Of the units sold at p, a share $r(p)$ are returned.

How $r(p)$ responds to price is central for what follows, so we unpack its derivative. Type i's return probability is

$$\phi_i(p) = \Pr(u < p - \kappa_i).$$

Willingness to pay settles who buys, but not who returns: buyers with the same v_i may still differ in κ_i and G_i. Averaging within WTP cells gives

$$\phi(v, p) \equiv \mathbb{E}\big[\phi_i(p) \;\big|\; v_i = v\big],$$

which depends on v whenever the joint distribution of (κ_i, G_i) varies across valuation levels. The aggregate return rate is the mean of these cell averages among those who buy at p,

$$r(p) = \frac{1}{D(p)} \int_p^\infty \phi(v, p)\, dF(v).$$

Differentiating and using $D'(p) = -f(p)$ splits the response into two terms,

$$r'(p) = \underbrace{\frac{1}{D(p)} \int_p^\infty \frac{\partial \phi(v,p)}{\partial p}\, dF(v)}_{\text{Behavioral channel}} + \underbrace{\frac{f(p)}{D(p)}\big(r(p) - \phi(p,p)\big)}_{\text{Selection channel}},$$

where $\phi(p,p)$ is the average return probability among types at the purchase margin $v_i = p$.

The first term is behavioral. Among consumers who remain in the market, a higher p raises the keep threshold and makes return strictly more likely. The second is selection. A higher price truncates the buyer pool at the margin, so composition changes whenever marginal buyers differ in return propensity from the average inframarginal buyer. For arbitrary (κ_i, G_i) that selection term is unsigned. Under the additive match $u = v_i + \varepsilon$ with ε and return hassle both independent of v_i, however, $\phi(v,p)$ decreases in v. Marginal buyers at $v_i = p$ then return more often than inframarginal buyers, so $\phi(p,p) \geq r(p)$ and selection is non-positive. The two forces therefore pull opposite ways. Which dominates is empirical, and we impose no sign restriction on $r'(p)$ below.

On the firm side the market is perfectly competitive. Producer j takes p as given, chooses output at cost $C_j(q_j)$, and incurs a net return handling cost h on each returned unit—gross handling cost minus scrap value. Net handling may be negative, but we maintain $p+h > 0$, which holds whenever scrap value does not exceed list price.

3 Consumer side

We turn first to consumer surplus and show that only the aggregates D and r enter. Write $cs(v,p)$ for expected surplus at list price p among types with willingness to pay v, so $cs(v,v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty cs(v,p)\,dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -cs(p,p)\,f(p) + \int_p^\infty cs_p(v,p)\,dF(v),$$

where $cs_p(v,p) \equiv \partial cs(v,p)/\partial p$. On the extensive margin, $cs(p,p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $cs_p(v,p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$cs_p(v,p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty cs_p(v,p)\,dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1 - r(p)\big)\,dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\,dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $cs_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once

(D, r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

4 Supply side

On the firm side, returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus net handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective margin*—is

$$\mu(p, h) \equiv p - r(p)(p + h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p, h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \; \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \;\; (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost. We assume each C_j has strictly increasing marginal cost, so $q_j'(\mu) > 0$ for active firms.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Under that assumption, $S'(\mu) > 0$. Market supply therefore rises with the effective margin, but not necessarily with list price. Firms care about μ, and $\mu(p, h)$ can fail to rise with p when refunds and handling eat into the list-price gain. Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\, dz. \tag{6}$$

At equilibrium $\mu = \mu(p, h)$, $PS = \int_0^{\mu(p,h)} S(z)\, dz$.

5 Failure of tax neutrality

With firms responding to the effective margin rather than the list price, a natural question is whether the classical result—that it does not matter which party remits a unit tax—survives. Weyl and Fabinger (2013) establish this neutrality without returns. With returns, the refund amount depends on the checkout price, and the answer is regime-dependent. To see the contrast, write the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p + h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p - \tau) - r(p - \tau)(p - \tau + h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p - \tau$ yields

$$r(p)(p + h) = r(p - \tau)(p - \tau + h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p + h) = r(p - \tau)(p - \tau + h)$.[4]

Tax neutrality breaks down because the private refund, and therefore the keep-or-return decision, depends on which party remits the tax. Under seller remittance the tax-inclusive checkout price is refunded. Under buyer remittance only the pre-tax list price is. In either case the remitter recovers none of τ from the government. Purchase is always based on checkout price p, but returns are evaluated at p under seller remittance and at $p - \tau$ under buyer remittance, so the effective margin generally differs and the two clearing conditions are not equivalent.

6 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p, h))$. Write $Q \equiv D(p)$. We characterize pass-through into list price for three shocks: a uniform one-dollar parallel shift in every firm's marginal production cost (indexed by c), a one-dollar increase in handling cost h, and a unit tax under seller and buyer remittance. Production cost shifts market supply at a given effective margin μ, while handling lowers $\mu(p, h)$ at a fixed list price p.

For production and handling, write the clearing condition as

$$m(p, x) \equiv D(p) - S(\mu(p, x), x) = 0, \qquad x \in \{c, h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

and expanding $m = D - S$ gives

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

[4] Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p - \tau$. That holds for all taxes only if $r(p)(p + h)$ is constant in posted list price, which requires $r(p) = C/(p + h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

hence

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{13}$$

The two shocks differ in which margin moves at fixed p or fixed μ,

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

The production term is $S_c = -S'(\mu)$ because a one-dollar parallel rise in every firm's marginal cost is equivalent, at fixed μ, to evaluating market supply at $\mu - 1$. Handling moves μ by only $r(p)$ per dollar relative to that unit production shift, so its numerator in (13) is the fraction $r(p)$ of the production numerator.

Turning to the unit tax, failure of tax neutrality requires separate checkout prices p^s and p^b under seller and buyer remittance. Write $\rho^s_\tau \equiv dp^s/d\tau$ and $\rho^b_\tau \equiv dp^b/d\tau$. At $\tau = 0$, (13) with $S_\tau = 0$ gives

$$\rho^s_\tau = \frac{-S'(\mu)\,\mu^s_\tau}{S'(\mu)\,\mu_p - D'(p^s)}, \qquad \rho^b_\tau = \frac{-S'(\mu)\,\mu^b_\tau}{S'(\mu)\,\mu_p - D'(p^b)}. \tag{15}$$

The regimes differ only through μ^s_τ and μ^b_τ at a fixed checkout price. At $\tau = 0$, $\mu^s_\tau = -1$. Under buyer remittance, τ enters the effective margin in the same way as $-p$: a tax increase lowers the pre-tax price entering $\mu(p-\tau, h)$ by one dollar, changing μ by $-\mu_p$. Hence $\mu^b_\tau = -\mu_p$ and $\rho^b_\tau = \mu_p\,\rho_c$. The sign of buyer-remitted pass-through is thus tied to that of seller-remitted pass-through by the sign of μ_p. When returns are absent ($r = 0$), $\mu_p = 1$ and the two pass-through rates coincide, recovering the classical neutrality benchmark. If $\mu_p < 0$, however, the two rates always carry opposite signs. This is a striking reversal of that neutrality intuition.

Result 2 (Pass-through into list price and remittance-dependent sign)**.** ***(i)*** *Production-cost pass-through is*

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}. \tag{16}$$

The remaining shocks connect to this object by

$$\rho_h = r(p)\,\rho_c, \qquad \rho^s_\tau = \rho_c, \qquad \rho^b_\tau = \mu_p\,\rho_c. \tag{17}$$

(ii) *If* $\mu_p < 0$*, then*

$$\operatorname{sgn}(\rho^b_\tau) = -\operatorname{sgn}(\rho_c)$$

provided $\rho_c \neq 0$*. A unit tax therefore moves the equilibrium checkout price in opposite directions depending on which side of the market remits it.*

The condition $\mu_p < 0$ requires the behavioural response of returns to price to be strong enough that the effective margin falls when p rises: from $\mu_p = 1 - r(p) - r'(p)(p + h)$, we have $\mu_p < 0$ exactly when $r'(p)(p+h) > 1 - r(p)$. In that case buyer remittance raises μ while seller remittance lowers it, shifting supply outward and inward respectively. The checkout price must therefore adjust in opposite directions to restore equilibrium. Which regime raises the price and which lowers it depends on the sign of ρ_c, which in turn hinges on the relative magnitudes of the supply and demand elasticities—a point developed in the next subsection.

6.1 Elasticity representation

The same pass-through formulas can be rewritten in elasticities, which makes the role of μ_p transparent. Write $\mu_p \equiv d\mu/dp$ for the response of the effective margin to list price, and define demand and supply elasticities with respect to list price as usual by

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp}\frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p\,\frac{p}{S(\mu)}.$$

Here $\varepsilon_S(p)$ is the observed elasticity of market supply with respect to the clearing price. Without returns, $\mu_p = 1$ and $\varepsilon_S > 0$. With returns, μ_p can become negative, in which case $\varepsilon_S < 0$ and a higher list price reduces the quantity firms are willing to supply.

Without returns, supply is a function of list price, so $S' = dS/dp$. A one-dollar cost increase cuts supply by S' units at the original price, creating an excess-demand gap of that size. Raising list price by one dollar closes the gap from both sides. Demand falls by $|D'|$ and supply recovers by S'. Closing a gap of size S' therefore requires a price rise of $S'/(S'+|D'|)$, so standard pass-through equals $\varepsilon_S/(-\varepsilon_D + \varepsilon_S)$.

With returns, supply is a function of the effective margin, so $S'(\mu) = dS/d\mu$ rather than dS/dp. A one-dollar production-cost increase is equivalent to lowering μ by one dollar, so at fixed list price supply falls by $S'(\mu)$ units, again creating an excess-demand gap of that size. Raising list price by one dollar closes the gap at rate $S'(\mu)\,\mu_p - D'(p)$, so

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}.$$

The cost shock lowers μ by one dollar, while a dollar of list price raises μ by μ_p. A higher μ_p means a one-dollar price increase restores more of the effective margin—and therefore more quantity supplied—than the one-dollar cost shock removes, so a smaller list-price movement closes the excess-demand gap. Measuring supply against list price through ε_S puts the factor $1/\mu_p$ in the numerator,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{18}$$

When $\mu_p < 0$, the supply curve expressed as a function of the list price p is locally decreasing. In that region the sign of ρ_c is no longer guaranteed positive. It depends on the relative magnitudes of ε_S and ε_D. This opens the door to the counterintuitive outcome that an increase in marginal cost reduces the equilibrium price.

Result 3 (Negative pass-through)**.** *We have $\rho_c < 0$ if and only if $\varepsilon_S < 0$ ($\mu_p < 0$) and $|\varepsilon_S| > -\varepsilon_D$.*

In words, negative pass-through occurs when supply slopes downward in list price—the behavioral return response dominates—and supply is steeper than demand. Under these conditions, a cost increase creates excess demand that can only be resolved by lowering the price, an outcome impossible in the classical model.

Proof. By (18),

$$\rho_c = \frac{\varepsilon_S/\mu_p}{-\varepsilon_D + \varepsilon_S}.$$

Since $\varepsilon_S = S'(\mu)\mu_p\,p/S(\mu)$, the terms ε_S and μ_p have the same sign, so $\varepsilon_S/\mu_p > 0$. Hence $\rho_c < 0$ if and only if

$$-\varepsilon_D + \varepsilon_S < 0,$$

which is equivalent to $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Because ε_S and μ_p have the same sign, $\varepsilon_S < 0$ is equivalent to $\mu_p < 0$. $\square$

The logic is as follows. With $\varepsilon_S < 0$ ($\mu_p < 0$), a higher list price lowers the quantity firms supply, so a leftward shift of supply—caused by a rise in c—creates excess demand at the initial p. Raising the price to eliminate the excess would further contract supply and worsen the imbalance; instead, the price must fall. A decline in p increases both the quantity demanded and, because supply slopes downward, the quantity supplied. When $|\varepsilon_S| > -\varepsilon_D$, the supply curve is steeper than the demand curve, meaning that the supply response to a price cut is larger than the demand response. The price decline therefore eliminates the excess demand and leads to a new equilibrium at a lower list price. This sign reversal cannot occur in the classical model, where $\mu_p = 1$ and supply is everywhere upward-sloping, highlighting a qualitative failure of the no-return benchmark.

Even when $\mu_p > 0$ and pass-through stays positive, the returns-adjusted formula differs from the no-return case whenever $\mu_p \neq 1$. Comparing the two signs the bias for an analyst who ignores returns. Whether μ_p exceeds one depends on how steeply the return rate falls with price. Differentiating $\mu = p - r(p)(p + h)$ gives

$$\mu_p = 1 - r(p) - r'(p)(p + h).$$

Write $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ for the elasticity of the return rate in list price. Then $\mu_p > 1$ if and only if $\varepsilon_r(p) < -p/(p + h)$. That is, selection must outweigh the behavioral rise in individual return probabilities by enough that aggregate r falls steeply in p. When instead the behavioral response dominates, ε_r is less negative (or positive) and $\mu_p < 1$.

Result 4. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\varepsilon_r(p) < -p/(p + h)$ and underestimates it otherwise.*

The practical value is that the bias can be signed without estimating the full returns-adjusted model. The return-rate elasticity and the ratio of handling to list price are enough. The economics behind the threshold is familiar from the selection and behavioral forces already in ε_r. When selection dominates, a higher list price lowers the return rate, so the refund-and-handling drag on the effective margin shrinks as price rises. The effective margin can then rise by more than one dollar per dollar of list price, and a smaller price movement closes the excess-demand gap than the classical formula predicts. Costly handling strengthens that channel and makes overestimation more likely. Otherwise, the effective margin rises by less than one dollar per dollar of list price, so list price must move by more than the classical formula predicts to close the gap, and the classical no-return formula underestimates pass-through.

7 Local incidence

Having established how prices respond to different shocks, we turn to local incidence: how each shock's pass-through ρ_x splits the resulting surplus change between consumers and producers. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. From the consumer-side analysis in Section 3, a small increase in any shifter x changes consumer surplus by

$$\frac{dCS}{dx} = -Q(1 - r)\rho_x, \tag{19}$$

where $\rho_x \equiv dp/dx$ is the pass-through of the shock into the list price.

Producer surplus is $PS(\mu, x) = \int_0^{\mu} S(z, x)\, dz$. Differentiating with respect to x and applying the equilibrium condition $S(\mu, x) = Q$ gives

$$\frac{dPS}{dx} = Q\frac{d\mu}{dx} + \Sigma_x,$$

where $\Sigma_x \equiv \int_0^{\mu}(\partial S(z,x)/\partial x)\,dz$ captures the direct shift in the area under the supply curve. Expanding the total derivative of the effective margin via the chain rule, $\frac{d\mu}{dx} = \mu_p\rho_x + \mu_x$, yields

$$\frac{dPS}{dx} = Q(\mu_p\rho_x + \mu_x) + \Sigma_x. \tag{20}$$

The local incidence ratio is $I_x \equiv (dCS/dx)/(dPS/dx)$, so

$$I_x = \frac{-Q(1-r)\rho_x}{Q(\mu_p\rho_x + \mu_x) + \Sigma_x}.$$

The components needed to evaluate I_x for each shock are reported in Table 1. For every shock the incidence ratio reduces to the same expression in ρ_c and μ_p, because the shock-specific pass-through and direct effects cancel when dCS/dx is divided by dPS/dx.

Shock x	ρ_x	μ_x	Σ_x	dCS/dx	dPS/dx	I_x
Production cost c	ρ_c	0	$-Q$	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Handling cost h	$r\rho_c$	$-r$	0	$-Q(1-r)r\rho_c$	$Qr(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Seller tax τ^s	ρ_c	-1	0	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Buyer tax τ^b	$\mu_p\rho_c$	$-\mu_p$	0	$-Q(1-r)\mu_p\rho_c$	$Q\mu_p(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$

Table 1: Pass-through, direct effects, surplus changes, and incidence ratio for each shock.

Result 5. *For all four shocks, the local incidence ratio is identical:*

$$I_x = \frac{(1-r)\rho_c}{1-\mu_p\rho_c} > 0. \tag{21}$$

Result 5 establishes that the incidence ratio is identical across all four shocks, but the algebra alone can obscure the deeper economic mechanism. The key to understanding this result is that every shock is a scaled perturbation of the same equilibrium geometry. Production cost and seller-tax shocks shift the effective margin μ down by one unit. Handling cost shifts it down by r units. A buyer tax shifts the mapping from list price to μ by μ_p units. Locally, the equilibrium condition $D(p) = S(\mu)$ is linear in the shock's direct effect on μ. The resulting supply shift is proportional to μ_x, and the price adjustment ρ_x that restores market clearing inherits that same proportionality.

Because both $S(\mu)$ and $PS(\mu)$ are locally linear in μ, the entire equilibrium response scales with the shock's effect on the effective margin. The shock's scalar therefore multiplies both consumer and producer surplus changes in the same proportion and cancels out in the incidence ratio. The shock determines how far the equilibrium moves. How that movement is split between consumers and producers depends on the local demand and supply slopes and on local μ_p.[5] Incidence is therefore fixed by market structure, not by which side of the market is hit first. What remains is why that common ratio is positive.

The denominator $1 - \mu_p\rho_c$ in (21) is the per-unit loss in producer surplus. A \$1 cost shock initially reduces producer surplus by Q. The equilibrium list-price rise ρ_c increases the effective

[5] Consumers care about the list-price change ρ_x. Producers recover only through the effective margin, which a list-price rise moves by $\mu_p\rho_x$. Thus μ_p governs how much of each dollar of p feeds into producer surplus.

margin per unit by $\mu_p\rho_c$. Because $dPS/d\mu = Q$ by the envelope theorem, this marginal gain translates into a total recovery of $Q\mu_p\rho_c$ in producer surplus. The net loss is therefore $Q(1-\mu_p\rho_c)$. The numerator $(1-r)\rho_c$ is the corresponding per-unit consumer loss on kept units. The incidence ratio compares these two per-unit burdens. A small denominator leaves consumers bearing almost the entire shock, while a negative denominator, $\mu_p\rho_c > 1$, means the recovery exceeds the cost, so producers are net gainers. This recovery factor can be written as $\mu_p\rho_c = \varepsilon^S/(\varepsilon^S - \varepsilon^D)$. This expression closely mirrors the pass-through formula, and now we argue that $\mu_p\rho_c > 1$ if and only if $\rho_c < 0$. The condition $\mu_p\rho_c > 1$ requires $\varepsilon^S < 0$, which implies $\mu_p < 0$, and hence $\rho_c < 0$. Conversely, $\rho_c < 0$ requires $\varepsilon^S < 0$ and $|\varepsilon^S| > -\varepsilon^D$, which gives $\mu_p\rho_c > 1$. Therefore, the condition governing producer gain from an increase in production cost is exactly the same as that governing negative pass-through.

At first glance, the price drop may seem puzzling. It raises the effective margin ($\mu_p < 0$ and $\rho_c < 0$) and necessarily more than offsets the original one-dollar cost increase. The key is that the price fall must do double duty to clear the market. A one-dollar cost increase shifts the supply curve left, creating an initial excess demand gap. To fill this gap, quantity supplied must increase. Since quantity supplied increases with the effective margin μ, restoring enough supply to cover the original gap already requires μ to rise by one dollar. But the price fall also increases quantity demanded, creating additional excess demand on top of the initial gap. To satisfy both the original supply shortfall and this new demand, supply must increase by more than the original shift, so the effective margin must rise by more than one dollar. Thus, $\mu_p\rho_c > 1$ is precisely the market-clearing requirement when pass-through is negative.

The preceding observation implies a striking result: the uniformly positive incidence ratio means consumer and producer surplus always move together, regardless of the sign of pass-through. The direction of that common movement nevertheless depends on both the shock type and the sign of pass-through. This yields a remarkable implication: a cost increase can be Pareto-improving. When $\rho_c < 0$, raising production costs, handling costs, or seller-remitted taxes lowers the list price, reduces return rates, and saves enough handling costs to more than offset the direct resource drain; both consumers and producers benefit. Under the same condition, buyer-remitted taxes reduce both surpluses, so the two tax regimes yield opposite welfare outcomes. This is a sharp violation of classical tax neutrality. When $\rho_c > 0$, all four shocks reduce both surpluses. Such gains are impossible in the classical model, where pass-through is always positive, underscoring how profoundly returns alter standard welfare analysis.

7.1 First-order private surplus and deadweight loss

Local incidence separates the burden between consumers and producers. Their joint payoff—private surplus—is the sum of consumer and producer surplus, $W \equiv CS + PS$. Summing (19) and (20) gives

$$\frac{dW}{dx} = -Q(1-r)\rho_x + Q(\mu_p\rho_x + \mu_x) + \Sigma_x = Q\big[\mu_x + \big(\mu_p - (1-r)\big)\rho_x\big] + \Sigma_x.$$

Substituting $\mu_p = 1 - r - r'(p)(p+h)$ directly yields

$$\frac{dW}{dx} = Q\big[\mu_x - r'(p)(p+h)\rho_x\big] + \Sigma_x, \tag{22}$$

with μ_x and Σ_x as in Table 1. The term $-Q\,r'(p)(p+h)\rho_x$ is new relative to the classical model. It captures the efficiency consequence of the price-induced change in the return rate. Each returned unit consumes real handling resources h, so a change in the return rate has a first-order effect on private surplus that is absent when returns are shut down.

Without returns, $r = 0$ and $r' = 0$, so (22) reduces to

$$\frac{dW}{dx} = Q\,\mu_x + \Sigma_x.$$

For a unit production-cost increase, the supply curve shifts vertically upward by one dollar at every quantity. The surplus loss is therefore the area of that shift—a rectangle of height one and width Q. Algebraically, private surplus is $W = CS(p) + PS(p - c)$. Differentiating, the price-change terms $-Q\rho_c$ and $+Q\rho_c$ cancel, leaving $dW/dc = -Q$: a first-order real resource loss of Q dollars.

For a unit tax the same cancellation gives $dW/d\tau = -Q$, but government revenue $G = \tau Q(\tau)$ is transferred rather than lost. Total surplus $V = W + G$ therefore satisfies $dV/d\tau = dW/d\tau + Q(\tau) + \tau Q'(\tau) = \tau Q'(\tau)$, which is zero at $\tau = 0$. Locally, a cost shock has a first-order private-surplus effect of $-Q$, while an infinitesimal tax has no first-order social loss—the dollar is lost in one case and merely transferred in the other.

Product returns overturn that clean private-versus-transfer contrast. The return-rate term in (22) is generally nonzero for every shock, so production-cost increases, handling-cost increases, and unit taxes under both remittance regimes all change private surplus at first order in a way that is not a pure transfer between buyers and sellers. We first track these private-surplus changes for all four shocks. For taxes, treasury revenue remains a transfer; we add $G = \tau Q(\tau)$ back when measuring deadweight loss below.

Plugging the values of μ_x and Σ_x from Table 1 into $dW/dx = Q\big[\mu_x - r'(p)(p+h)\rho_x\big] + \Sigma_x$, and using the pass-through links from Result 2, yields

$$\frac{dW}{dc} = -Q\,\Lambda, \qquad \frac{dW}{dh} = -Q\,r\,\Lambda, \qquad \frac{dW}{d\tau^s} = -Q\,\Lambda, \qquad \frac{dW}{d\tau^b} = -Q\,\mu_p\,\Lambda, \tag{23}$$

where the common factor is defined as

$$\Lambda \equiv 1 + r'(p)(p+h)\,\rho_c. \tag{24}$$

Private surplus moves because consumers change either how much they buy or whether they return what they bought. The unit term in Λ is the mechanical effect on the purchase margin that remains even if the return rate does not respond to price. The term $r'(p)(p+h)\rho_c$ is the extra efficiency cost on the keep-or-return margin after the price moves: each change $r'(p)\rho_c$ in the return rate costs $p+h$ per unit sold. We therefore decompose the private-surplus changes in (23) into an ex-ante loss on the purchase margin and an ex-post loss on the keep-or-return margin. That split is simplest for a production-cost or seller-tax shock, where $dW/dx = -Q\Lambda$; handling and buyer-remitted taxes rescale both pieces by r and μ_p, respectively.

Freeze the aggregate return rate at its initial value $r_0 \equiv r(p_0)$. In the resulting ex-ante economy the effective margin is (4) with $r(p)$ replaced by that constant,

$$\mu^{\mathrm{EA}}(p) \equiv p - r_0(p+h). \tag{25}$$

Applying (22) then yields $dW/dx\big|_{\mathrm{EA}} = -Q$, the classical rectangle from changing gross quantity alone. The ex-post change is that residual, $dW/dx\big|_{\mathrm{EP}} \equiv dW/dx - dW/dx\big|_{\mathrm{EA}}$. Since $dW/dx = -Q\Lambda$ for these shocks,

$$\left.\frac{dW}{dx}\right|_{\mathrm{EP}} = -Q\,r'(p)(p+h)\rho_c. \tag{26}$$

Result 6 (Ex-ante and ex-post decomposition)**.** *For a unit production-cost or seller-tax shock,*

$$\frac{dW}{dx} = -Q - Q\,r'(p)(p+h)\rho_c. \tag{27}$$

The first term is the ex-ante loss on the purchase margin. The second is the ex-post loss on the keep-or-return margin. Each unit switched from keep to return costs $p + h$. For a handling shock the two losses become $-Qr$ and $-Qr\, r'(p)(p+h)\rho_c$. For a buyer-remitted tax they become $-Q\mu_p$ and $-Q\mu_p\, r'(p)(p+h)\rho_c$, so the ex-ante loss is not $-Q$ whenever $\mu_p \neq 1$.

Because consumer and producer surplus always move together, Λ has the same sign as ρ_c. The no-return formula retains only the ex-ante rectangle $-Q$, so the comparison with that benchmark is governed only by the ex-post term $-Q\, r'(p)(p+h)\rho_c$: that term may amplify the rectangle, partially offset it, or reverse the private-surplus change.

When $\rho_c > 0$ and $\mu_p > 0$, a cost increase raises the list price. If $r'(p) > 0$, that price rise raises the return rate and adds handling costs, so the ex-post loss amplifies the ex-ante rectangle and $\Lambda > 1$. If instead $r'(p) < 0$, the return rate falls and the ex-post term partially offsets the ex-ante loss, yielding $0 < \Lambda < 1$. The private-surplus loss $Q\Lambda$ is larger when $r'(p)$ or h is large, because each dollar of pass-through then wastes more $p + h$ per unit sold.

When $\rho_c > 0$ but $\mu_p < 0$ (so $|\varepsilon_S| < -\varepsilon_D$), supply slopes downward in list-price space and pass-through remains positive only because demand is more price-responsive than supply. The return-rate response must already be steep—$\mu_p < 0$ requires $r'(p)(p+h) > 1-r$—so even a modest ρ_c can make the ex-post loss large relative to the ex-ante rectangle and push Λ well above one.

When $\rho_c < 0$, a production-cost or seller-tax increase lowers the list price and, with it, the return rate. Units that would have been returned are kept instead; each such switch saves $p + h$, and the ex-post term becomes a gain. Those savings can exceed the ex-ante resource cost, so private surplus rises by $Q|\Lambda|$ and a Pareto improvement is possible. The gain is larger when $r'(p)(p+h)\,|\rho_c|$ is large.

Result 7 (Bias from ignoring returns in first-order private surplus)**.** *Consider a unit production-cost or seller-tax increase. Relative to the correct returns-adjusted change $dW/dc = -Q\Lambda$, the no-return formula $dW/dc = -Q$*

(i) underestimates the private-surplus loss if and only if $\rho_c > 0$ and $r'(p) > 0$ (hence $\Lambda > 1$);

(ii) overestimates the private-surplus loss if and only if $r'(p) < 0$ (which implies $\mu_p > 0$ and hence $\rho_c > 0$, so $0 < \Lambda < 1$);

(iii) predicts a loss of Q when private surplus in fact rises if and only if $\rho_c < 0$, in which case the rise is $Q|\Lambda|$.

Under buyer remittance both margins scale by μ_p, so the comparison with $-Q$ mixes an ex-ante factor μ_p and an ex-post term scaled by μ_p.

Corollary 1 (Bias under buyer remittance)**.** *Relative to the no-return prediction $dW/d\tau = -Q$, a unit buyer-remitted tax has true change $-Q\mu_p\Lambda$.*

(i) If $\mu_p > 0$ (which implies $\rho_c > 0$ and hence $\Lambda > 0$), both components keep the seller-remittance signs but scale with μ_p. The no-return formula underestimates the private-surplus loss if and only if $\mu_p\Lambda > 1$ and overestimates it if and only if $0 < \mu_p\Lambda < 1$.

(ii) If $\mu_p < 0$, the ex-ante piece flips relative to $-Q$ and the ex-post piece flips with it. The no-return formula predicts a loss of Q, whereas private surplus rises by $Q\,|\mu_p|\Lambda$ if and only if $\rho_c > 0$ and falls by $Q\,|\mu_p\Lambda|$ if and only if $\rho_c < 0$.

Appendix A restates these cases in the return-rate elasticity $\varepsilon_r(p)$ and the observed supply and demand elasticities $\varepsilon_S(p)$ and $\varepsilon_D(p)$—the form most useful for empirical checks.

For tax shocks, social surplus adds back treasury revenue. Because the remitter recovers none of τ from the government on a return, revenue is $G = \tau Q(\tau)$ under both remittance regimes, with Q the equilibrium gross quantity at that tax. Write $V = W + G$. Differentiating the product gives $dG/d\tau = Q + \tau Q'$, so at $\tau = 0$

$$\frac{dV}{d\tau} = \frac{dW}{d\tau} + Q.$$

Using (23),

$$\frac{dV}{d\tau^s} = -Q(\Lambda - 1) = -Q\, r'(p)(p+h)\rho_c, \qquad \frac{dV}{d\tau^b} = Q\big(1 - \mu_p \Lambda\big). \tag{28}$$

The first-order deadweight loss of an infinitesimal seller-remitted tax is therefore only the ex-post private-surplus piece: the revenue term $+Q$ cancels the ex-ante rectangle, and the $\tau Q'$ contribution vanishes at $\tau = 0$. Without returns, $\Lambda = 1$ and that loss is zero. Under buyer remittance the private change is scaled by μ_p, so $+Q$ does not cancel one-for-one unless $\mu_p = 1$. Cost and handling shocks have $G = 0$, so their social first-order effects coincide with (23).

8 Global Incidence

Integrating the local derivatives $\frac{dCS}{dx}$ and $\frac{dPS}{dx}$ listed in Table 1 from x_0 to x_1 gives the global surplus changes ΔCS and ΔPS for any shock $x \in \{c, h, \tau^s, \tau^b\}$. The global incidence ratio $I_{x_0}^{x_1} \equiv \Delta CS/\Delta PS$ (when $\Delta PS \neq 0$) follows from the local relation $dCS/dx = I(x)\, dPS/dx$:

$$I_{x_0}^{x_1} = \int_{x_0}^{x_1} I(x)\, \omega(x)\, dx, \tag{29}$$

where $\omega(x) \equiv (dPS/dx)/\Delta PS$. Although $I(x) > 0$ at every point, the global ratio need *not* be positive: if dPS/dx changes sign along the path, the weights $\omega(x)$ take both signs, and the weighted average can become negative. Hence the classical sign-preserving property of local incidence does not extend globally.

A Welfare bias in primitives

Recall $\varepsilon_r(p) \equiv p\,r'(p)/r(p)$ and $\mu_p = 1 - r(p) - r'(p)(p+h)$. Then $r'(p)(p+h) = \varepsilon_r(p)\,r(p)\,(p+h)/p$, so

$$\mu_p > 0 \iff \varepsilon_r(p) < \frac{p}{p+h}\,\frac{1-r(p)}{r(p)}, \tag{30}$$

$\mu_p < 0$ is the reverse strict inequality, and ε_r has the same sign as r' (throughout, $r(p) \in (0,1)$).

Result 7 becomes:

(i) underestimation iff $\varepsilon_r(p) > 0$ and either $\mu_p > 0$, or both $\mu_p < 0$ and $|\varepsilon_S| \le -\varepsilon_D$;

(ii) overestimation iff $\varepsilon_r(p) < 0$;

(iii) wrong sign (predicted loss, actual gain) iff $\mu_p < 0$ and $|\varepsilon_S| > -\varepsilon_D$.

Proof. By Result 7, underestimation is $\rho_c > 0$ with $r' > 0$, overestimation is $r' < 0$, and the wrong-sign case is $\rho_c < 0$. Since ε_r has the same sign as r', write these in ε_r. Result 3 translates the sign of ρ_c into conditions on μ_p and $|\varepsilon_S| \gtreqless -\varepsilon_D$: with $\varepsilon_r > 0$ this is (i), and $\rho_c < 0$ is (iii). For (ii), $\varepsilon_r < 0$ forces $\mu_p > 1 - r > 0$ and hence $\rho_c > 0$, so $0 < \Lambda < 1$. $\square$

Corollary 1 becomes:

(i) If $\mu_p > 0$, underestimation iff $\dfrac{\varepsilon_S - \varepsilon_D}{\varepsilon_D} > \varepsilon_r(p)\dfrac{p+h}{p}$, and overestimation iff $\dfrac{\varepsilon_S - \varepsilon_D}{\varepsilon_D} < \varepsilon_r(p)\dfrac{p+h}{p}$;

(ii) If $\mu_p < 0$, surplus rises iff $|\varepsilon_S| \le -\varepsilon_D$ and falls iff $|\varepsilon_S| > -\varepsilon_D$.

Proof. Under $\mu_p > 0$ we have $\varepsilon_S > 0$, so $\varepsilon_S - \varepsilon_D > 0$, and also $\rho_c > 0$, $\Lambda > 0$. Write $\mu_p = 1 - r(p) - r'(p)(p+h)$ and $\Lambda = 1 + r'(p)(p+h)\rho_c$. Using $\mu_p\rho_c = \varepsilon_S/(\varepsilon_S - \varepsilon_D)$,

$$\begin{aligned}
\mu_p\Lambda - 1 &= \mu_p\big(1 + r'(p)(p+h)\rho_c\big) - 1 \\
&= \mu_p - 1 + r'(p)(p+h)\,(\mu_p\rho_c) \\
&= -r(p) - r'(p)(p+h) + r'(p)(p+h)\,\frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} \\
&= -r(p) + r'(p)(p+h)\,\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D} \\
&= -r(p)\left(1 - \varepsilon_r(p)\,\frac{p+h}{p}\,\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D}\right).
\end{aligned}$$

Hence $\mu_p\Lambda > 1$ iff $\varepsilon_r(p)\frac{p+h}{p}\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D} > 1$. Since $\varepsilon_D/(\varepsilon_S - \varepsilon_D) < 0$, multiplying through reverses the inequality and yields the underestimation cut in (i). The overestimation cut is the reverse; under $\mu_p > 0$ and $\Lambda > 0$ it is exactly $0 < \mu_p\Lambda < 1$. Part (ii): under $\mu_p < 0$, surplus rises iff $\rho_c > 0$ and falls iff $\rho_c < 0$, which is Result 3. Regions may be read in ε_r via (30). $\square$

References

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